

Emilie Pereira

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PROFESSIONAL SUMMARY

Mechanical Engineering student at the University of Maryland with hands-on experience in CAD modeling, 3D printing, and physical prototyping. Built and designed functional systems across multiple projects, from automated control systems to fabricated scale models. Eager to contribute to product design teams through strong design fundamentals and a proven ability to bring concepts to physical form.

EDUCATION

University of Maryland, College Park

Bachelor of Science in Mechanical Engineering.

Expected May 2029

Montgomery College

Part-Time General Education Coursework. GPA: 4.0/4.0.

Rockville, MD

May 2025

James Hubert Blake High School

Honors, Advanced, and AP Coursework. GPA: 3.92/4.0.

Colesville, MD

June 2025

PROJECTS

3D Printed Solar-Lit Home

- Independently modeled all components in Onshape over one to two months; 3D printed the full house structure in a single print, framed the interior, added a half-roof to mount a solar panel, and wired the panel to power LED lighting throughout.

Hydraulic Arm

- Designed and built a hydraulic arm as an individual entry in a ~10-person competition, earning first place and a \$50 cash prize.

Fire Suppression System

- Designed and programmed a small-scale automated fire detection and suppression system using an Arduino, heat sensor, smoke sensor, and water pump; furnished the interior with mini furniture to simulate a real environment.

Wooden Tower Strength Challenge

- Constructed a thin wooden tower weighing 0.195 lbs that withstood approximately 90 lbs of applied load during testing.

EXPERIENCE

Team Member

June 2025 – August 2026

Playa Bowls

Silver Spring, MD

- Managed cash, register, inventory, and open/close duties alongside food prep during high-volume weekend rushes.
- Resolved customer complaints in person and by phone, de-escalating concerns and leaving customers satisfied.
- Trained new staff on store procedures and upheld a clean, efficient work environment.

Secretary

November 2023 – June 2024

Pereira Construction

Silver Spring, MD

- Managed daily office operations including scheduling meetings, coordinating calendars, and organizing files.
- Handled incoming calls, emails, and correspondence, routing communications to appropriate departments.

Teacher Assistant Volunteer

September 2021 – June 2022

St. Patrick's Catholic Church

Rockville, MD

- Supervised children and fostered a safe, positive learning environment in a classroom setting.

ACTIVITIES & LEADERSHIP

Ambassadors Club

President

- Led approximately 30 members in supporting school events and outreach; coordinated logistics including timing and transportation for club-assisted school events.

Girls Lacrosse

Team Captain

- Led and coordinated a team of athletes through competitive seasons, fostering positive team dynamics and sportsmanship.

National Honor Society (Math and English)

Member

- Tutored peers in math and English and contributed to community service projects.

SKILLS

Design and Fabrication: CAD (Onshape, Creo Parametric), 3D Printing, Hydraulic Systems, Robotics and Mechanical Assembly

Programming: Embedded and Microcontroller Programming (Arduino)

Professional: Communication, Leadership, Teamwork